A1. Gym Membership Foundation & Batch Sign-up

Category B Assignment - Problem 1 (Basic)

Approach

The constructor owns the member-ID validation. Batch signup constructs each entry and counts constructor failures with try/catch; it does not pre-check IDs. PremiumMember uses direct single inheritance and super(...).

Java Solution

```
class GymMember {
    protected final String memberId;
    protected final int monthlyFee;
    private int sessionsAttended;

    public GymMember(String memberId, int monthlyFee) {
        if (memberId == null || memberId.trim().length() < 4)
            throw new IllegalArgumentException("Invalid member ID");
        if (monthlyFee <= 0)
            throw new IllegalArgumentException("Monthly fee must be positive");
        this.memberId = memberId;
        this.monthlyFee = monthlyFee;
    }
    public void attendSession() { sessionsAttended++; }
    public int getSessionsAttended() { return sessionsAttended; }
}

class PremiumMember extends GymMember {
    private final String trainerName;
    public PremiumMember(String memberId, int monthlyFee, String trainerName) {
        super(memberId, monthlyFee);
        this.trainerName = trainerName;
    }
}

class BatchSignUp {
    static String signUpBatch(String[] memberIds, int monthlyFee) {
        int signedUp = 0, rejected = 0;
        for (String id : memberIds) {
            try { new GymMember(id, monthlyFee); signedUp++; }
            catch (IllegalArgumentException ex) { rejected++; }
        }
        return "Signed Up: " + signedUp + " | Rejected: " + rejected;
    }
    public static void main(String[] args) {
        System.out.println(signUpBatch(
            new String[]{"MEM1", "GM1", "MEM2", " ", "MEM3"}, 1000));
    }
}
```

Sample Output

```
Signed Up: 3 | Rejected: 2
```